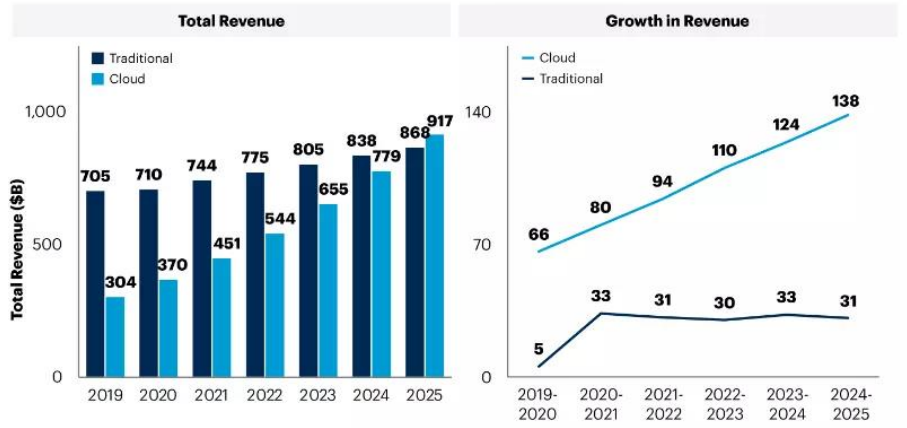
Public Cloud Services
IaC

IaC in Practice

Software eats the world…

Figure 1: Sizing Cloud Shift, Worldwide, 2019 – 2025

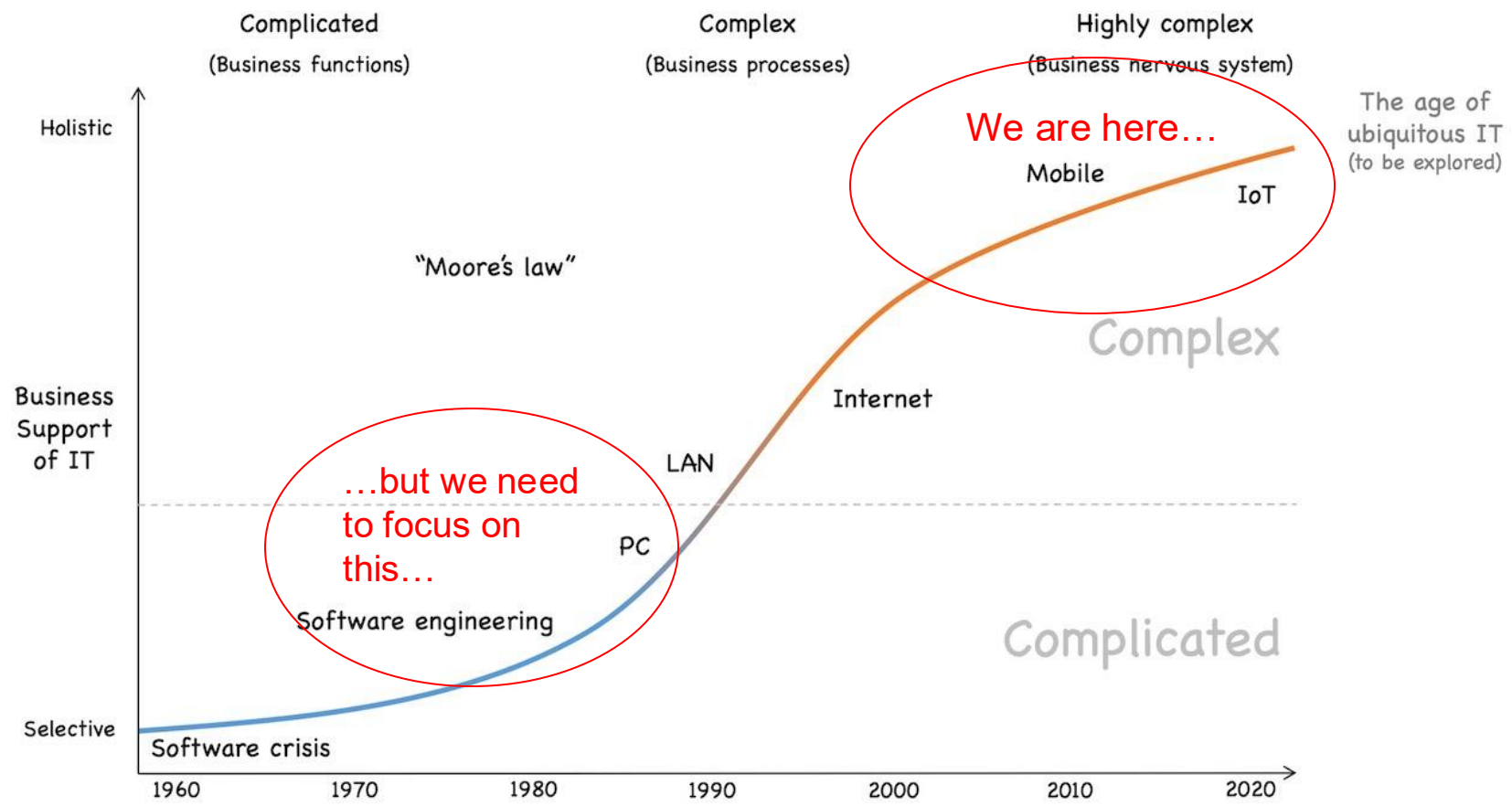


Source: Gartner, February 2022

Workloads are shifting towards virtualized environments

→ Need for software engineering practices kicks in…

Problem Field



Modules for Softwaredevelopment	Modules for Softwareoperations
>20	<10
Programming Languages, Technologies, etc.	Processes, System Engineering, -management

Iron Age VS Cloud Age

Iron Age	Cloud Age
Physical hardware	Virtualized resources
Provisioning takes weeks	Provisioning takes minutes
Manual processes	Automated processes

Iron Age	Cloud Age
Cost of change is high	Cost of change is low
Changes represent failure (changes must be "managed," "controlled")	Changes represent learning and improvement
Reduce opportunities to fail	Maximize speed of improvement
Deliver in large batches, test at the end	Deliver small changes, test continuously
Long release cycles	Short release cycles
Monolithic architectures (fewer, larger moving parts)	Microservices architectures (more, smaller parts)
GUI-driven or physical configuration	Configuration as Code

5 Principles of Infrastructure in the Cloud Age

Assume that Systems are not reliable

- Complex Systems → multiple Failure Causes
- Upgrade/Update/Resize need reprovisioned Systems (or at least reboot)
- Systems need to be abstracted from underlaying hardware

Make everything reproducible

- If you build a system, be sure you can rebuild it
- Stage-Parity: Test/Prod should be the same
- Enable horizontal scaling by supporting scale out

Create disposable systems

- Cloud infrastructure is as abstract as Software, it may be deleted as Software
- Infrastructure is not static but elastic. Add, alter and remove are natural operations

5 Principles of Infrastructure in the Cloud Age

Minimize Variations

- Each variation generates manual work or technical debt
- Operating 1000 services from the same kind is easier than operating 50 services of different kinds
  - Keep setup,OS, Architecture, Deployment Kind similar / the same

The process of creation / deployment should be common work

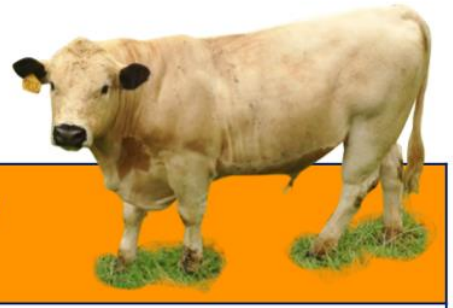
- Reproducible actions become robust, traceable, secure, tested, …
- Bus-Faktor > 1
- Easy/Flexible Stage Parity
- Scripting, Scripting, Scripting

Pet VS Cattle

Pets

Legacy Infrastructure
Pets are given names like grumpycat.petstore.com They are unique, lovingly hand raised and cared for When they get ill, you nurse them back to health
Infrastructure is a permanent fixture in the data center
Infrastructure takes days to create, are serviced weekly, maintained for years, and requires migration projects to move
Infrastructure is modified during maintenance hours and generally requires special privileges such as root access
Infrastructure requires several different teams to coordinate and provision the full environment
Infrastructure is static, requiring excess capacity to be dormant for use during peak periods of demands
Infrastructure is an capital expenditure that charges a fix amount regardless of usage patterns

Cattle

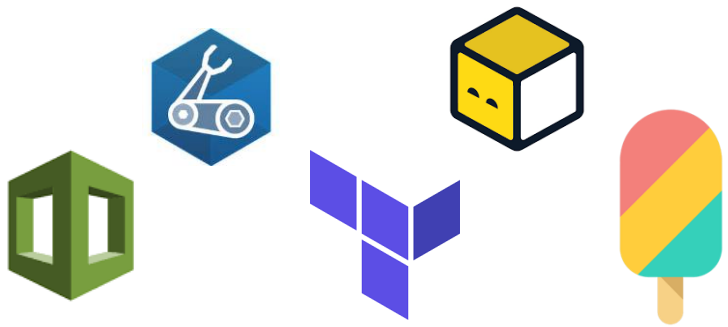


Cloud-Friendly Infrastructure
Cattle are given numbers like 10200713.cattlerancher.com They are almost identical to other cattle When they get ill, you replace them and get another
Infrastructure is stateless, ephemeral, and transient
Infrastructure is instantiated, modified, destroyed and recreated in minutes from scratch using automated scripts
Infrastructure uses version-controlled scripts to modify any service without requiring root access or privileged logins
Infrastructure is self-service with the ability to provision computing, network and storage services with a single click
Infrastructure is elastic and scales automatically, expanding and contracting on-demand to service peak usage periods
Infrastructure is a operating expenditure that charges only for services when they are consumed

Infrastructure as Code

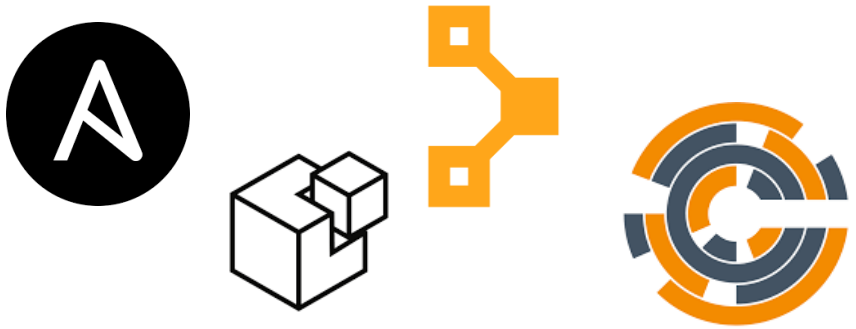
Infrastructure Provisioning

- Speaks with API of infrastructure provider
- Provisions ”atomic” elements / services
- Seldomly adaption of existing services, re-creation instead
  - Be aware if you have data, data may be deleted of existing services
- Needs to handle state



Configuration Management

- Speaks with (IaaS)-servers / localhost
- Adapt instances and services
- Mostly adaption of existing services, re-creation only with remote services
  - Quite robust when it comes to existing services
- State is retrieved at runtime



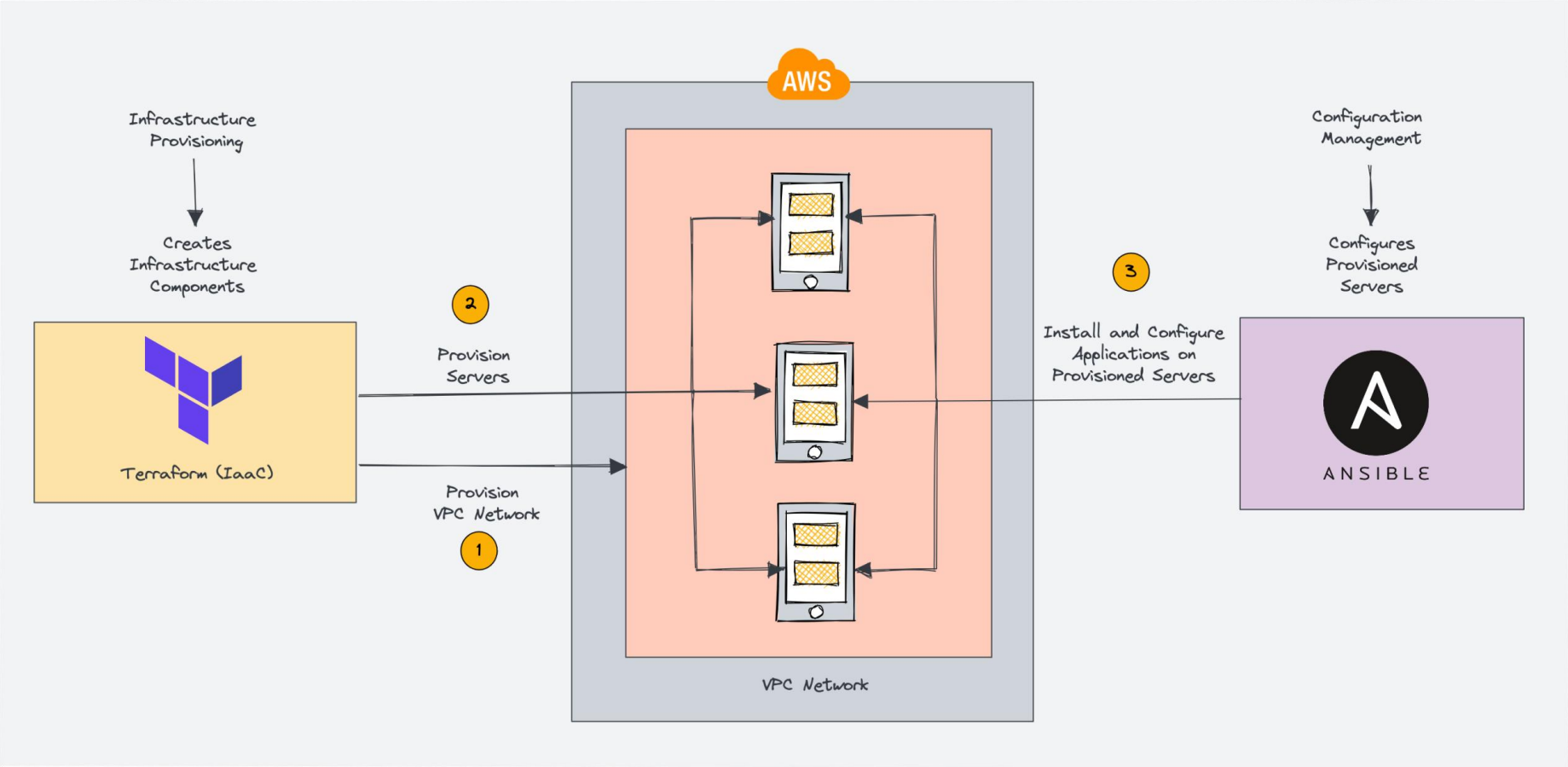
Survey…

Survey

Company Name	Tool Used	Main Use Case	Satisfaction Level (1-5)	Key Features Used	Challenges Faced
Company A	Ansible	Configuration Management	4	Playbooks, Roles	Complex Playbook Management
Company B	Terraform	Infrastructure Provisioning	5	HCL, State Management	Initial Setup Complexity
Company C	Both	Mixed (Infra & Configuration)	3	Ansible for Config, Terraform for Infra	Integration Issues, Learning Curve
Company D	Terraform	Cloud Infrastructure Automation	4	Modules, State Files	Handling Multi-Cloud Environments
Company E	Ansible	Application Deployment	4	YAML Playbooks, Roles	Scaling Issues
Company F	Terraform	Multi-Cloud Resource Management	5	Providers, Resource Plans	Terraform State File Management
Company G	Ansible	Orchestration	3	Ad-Hoc Commands, Roles	Complex Dependencies Management
Company H	Both	General Automation	4	Combination of Both Tools	Complexity in Managing Both Tools
Company I	Terraform	Infrastructure as Code	5	Modules, Workspaces	Initial Learning Curve
Company J	Ansible	System Configuration	4	Playbooks, Inventory Files	Handling Large Playbooks

https://rjpn.org/jetnr/papers/JETNR2308001.pdf

Infrastructure Provisioning ♡ Configuration Management

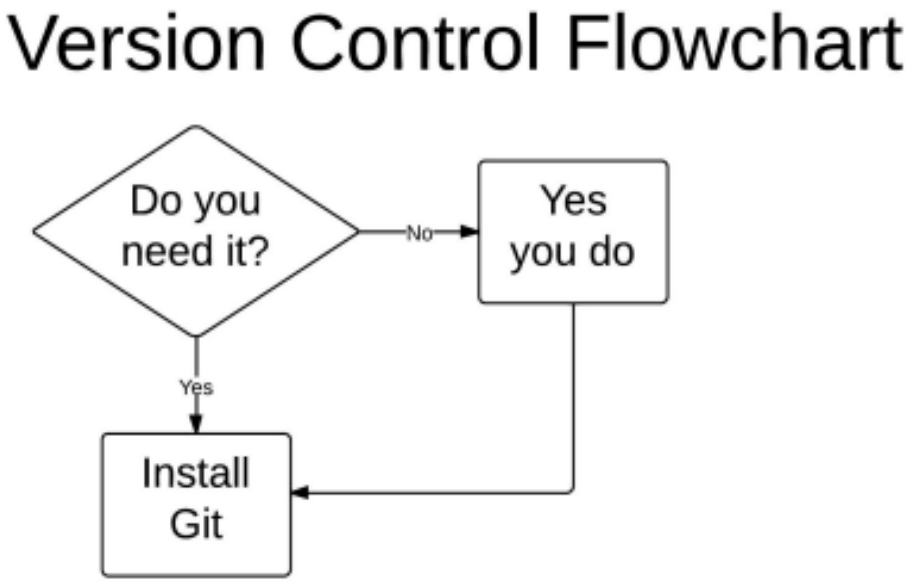


IaC is Source code

Sourcecode-Mgmt is a must have:

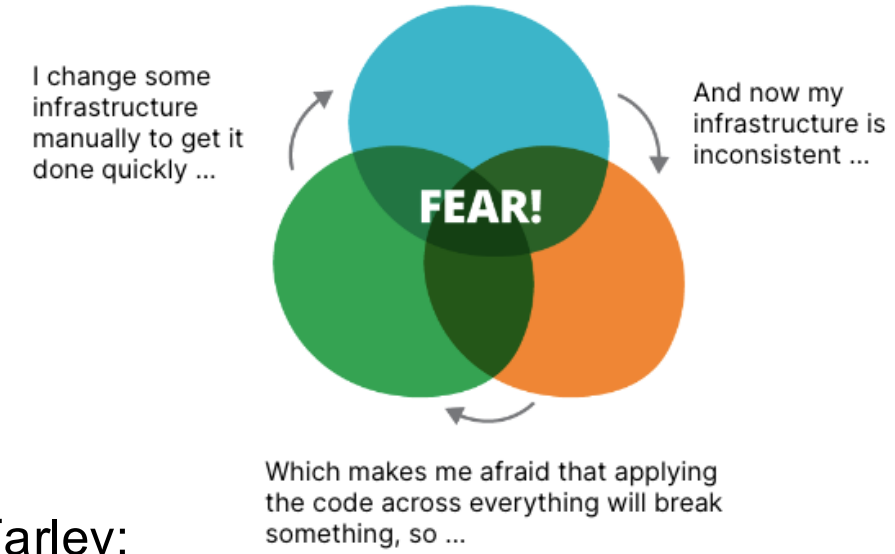
All configurations need to be under source control, even locally (e.g. etckeeper)

- Linting:
  - Syntaxchecks
  - Existieren für fast alle Sprachen
- Testing
  - Sematicchecks
- Auditing
  - Merge Request
- Versioning
  - Reverting



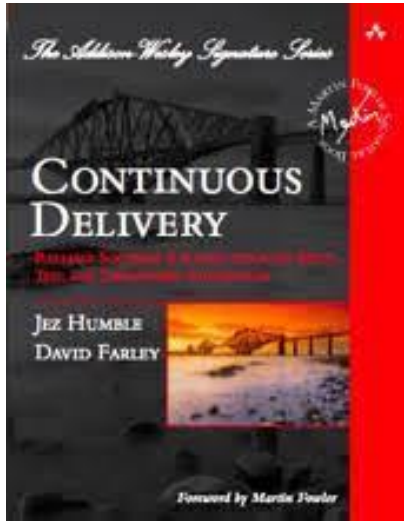
Pitfalls:
Fear the Fear Spiral

Infrastructure follows Code…including removal:
→ Lifecycle (including removal) is part of the game



Principles of Software Delivery from Jez Humble and David Farley:

1. *Create a Repeatable, Reliable Process for Releasing Software*
2. *Automate Almost Everything*
3. *Keep Everything in Version Control*
4. *If It Hurts, Do It More Frequently, and Bring the Pain Forward*
5. *Build Quality In*
6. *Done Means Released*
7. *Everybody Is Responsible for the Delivery Process*
8. *Continuous Improvement*

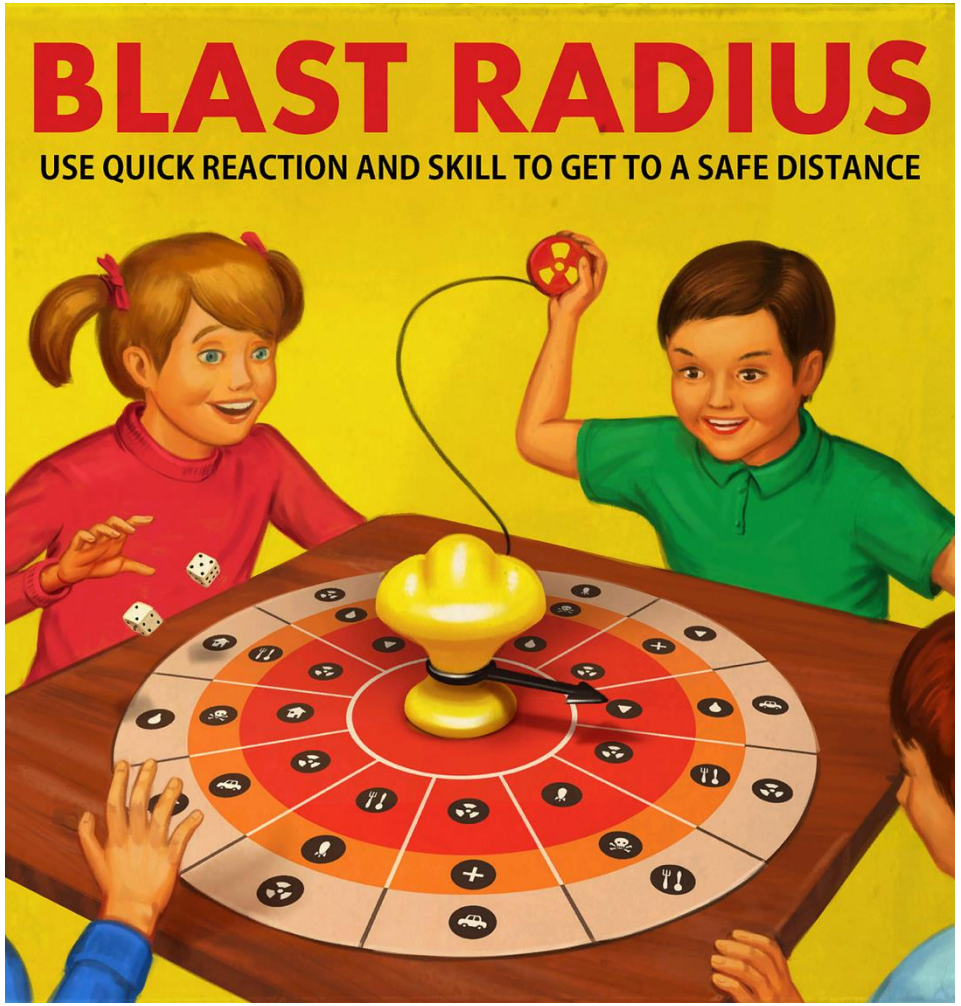


Pitfalls:
Mind the Blast Radius

IaC influences applications:

→ We are on the bottom of the stack, if we are having failures here, all upper layers are affected

- Reducing side effects by defining fixed defined accounts / subscriptions / projects / namespaces / resource groups
- Use testing / staging to become aware of blast radius
- Use tagging / conventions of resources



Pitfalls:
How should infrastructure be defined in Sourcecode?

```
stack-tool create-virtual-server \
  --name=my-server \
  --os_image=ubuntu-22.10 \
  --memory=4GB \
  --disk=80GB
```

Pitfalls:
How should infrastructure be defined in Sourcecode?

```
stack-tool find-virtual-server --name=my-server
if [ "$?" = "1" ] ; then
  echo "Creating a new server"
  stack-tool create-virtual-server \
    --name=my-server \
    --os_image=ubuntu-22.10 \
    --memory=4GB \
    --disk=80GB
else
  echo "Server already exists, not creating a new one"
fi
```

Pitfalls:
How should infrastructure be defined in Sourcecode?

```
stack-tool find-virtual-server --name=my-server --print-id > saved-server-id
if [ "$?" = "1" ] ; then
  echo "Creating a new server"
  stack-tool create-virtual-server \
    --name=my-server \
    --os_image=ubuntu-22.10 \
    --memory=8GB \
    --disk=80GB
else
  echo "Server already exists, not creating a new one"
fi
CURRENT_RAM=`stack-tool get-virtual-server-info \
  --id=$(cat saved-server-id) \
  --attribute=memory`
if [ "${CURRENT_RAM}" != "8GB" ] ; then
  stack-tool modify-virtual-server \
    --id=$(cat saved-server-id) \
    --memory=8GB
fi
```

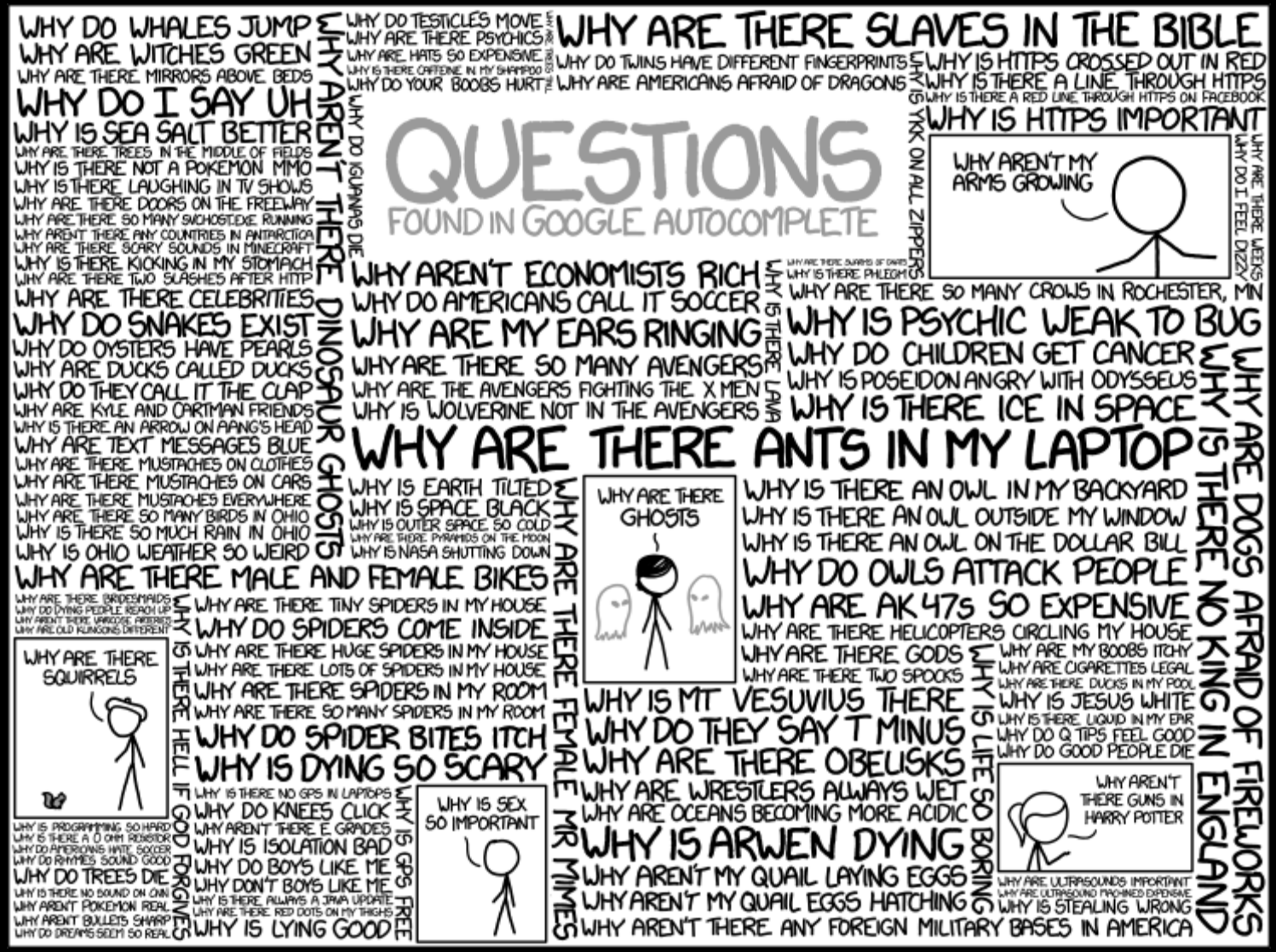
Pitfalls:
How should infrastructure be defined in Sourcecode?

Imperative	Declarative
Explicite Instructions	Describe the outcome
Flexible	Domain Specific
High levels of control	Difficult to optimize
The system is stupid, you are smart	The system is smart, you don't care

```
virtual_machine:
  name: my_application_server
  source_image: 'base_linux'
  cpu: 2
  ram: 2GB
  network: private_network_segment
  provision:
    provisioner: servermaker
    role: tomcat_server
```

Summary

- Declarative Languages to the rescue
  - Not caring how but what
  - Hard to debug
- Different tools for different levels
- Infrastructure Provisioning vs Configuration Management
  - Do not seek for the “one” solution
  - Try to combine best of worlds
  - Do not use different tools for same purpose


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
WHY AREN'T THERE VARICOSE ARTERIES
WHY ARE OLD KLINGONS DIFFERENT
WHY ARE THERE SQUIRRELS
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY ARE THERE SWARMS OF DAVES
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY IS SEX SO IMPORTANT
WHY DO IGUANAS DIE
WHY AREN'T THERE DINOSAUR GHOSTS
WHY IS THERE LAVA
WHY ARE THERE FEMALE MR MIMES
WHY IS GPS FREE
WHY IS LIFE SO BORING
WHY IS THERE NO KING IN ENGLAND
WHY ARE DOGS AFRAID OF FIREWORKS
WHY DO I FEEL DIZZY
WHY ARE THERE WEEKS
WHY IS YYK ON ALL ZIPPERS
WHY DO FORGIVES